CANTEEN FOOD WASTE PREDICTION & TRACKER

Using Arrays, Sorting and Searching
IT23302 – Data Structures Project

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1. Introduction

Every day a canteen cooks a large quantity of food, and a part of it is left unused. Since the quantities prepared, eaten and wasted are rarely written down, the same mistakes are repeated daily. This project is a C program that records these quantities, shows where the waste happens and suggests how much to cook next time.

2. Aim of the Project

- To keep a proper record of food prepared and food consumed each day.
- To identify the items that create the most waste.
- To help the canteen cook the right quantity and save cost.
- To practise Arrays, Sorting and Searching through a real-life problem.

3. Concepts Used

- **Arrays:** Hold the daily records such as item name, date, prepared, consumed and wasted quantity.
- **Sorting:** Orders items by waste (highest first) or by date, using Bubble Sort or Selection Sort.
- **Searching:** Looks up a record by item name or date using Linear Search, and Binary Search on date-sorted data.

4. Key Features

- Waste is calculated automatically as prepared minus consumed.
- Waste percentage is shown for every item.
- Next day's quantity is predicted from the average of earlier days.
- A ranked list of the most wasted items is displayed.
- Records are saved in a file and reloaded when the program starts.

5. Program Flow

1) Start the program and choose an option from the menu.
2) Enter the item, date, quantity prepared and quantity consumed.
3) The program calculates the waste and stores the record in the array.
4) Sort the records by waste or by date when required.
5) Search for any record by item name or date.
6) View the predicted quantity and the summary report.
7) Save the records to the file and exit.

6. Functions of the Program

- **addRecord():** Reads a new entry and updates the array.

- **calculateWaste():** Finds the waste quantity and waste percentage.
- **predictQuantity():** Suggests the quantity to prepare next day.
- **sortRecords():** Sorts the array by waste or by date.
- **searchRecord():** Finds a record by item name or date.
- **showReport():** Prints the daily and weekly summary.
- **saveToFile() / loadFromFile():** Keeps data safe between runs.

7. Expected Benefits

- Less food is wasted and the canteen saves money.
- Staff can see at a glance which items need smaller quantities.
- Records are kept neatly and can be found quickly.
- The project shows how basic data structures solve everyday problems.